nature portfolio

Peer Review File

Targeting climate finance for global forests

Corresponding Author: Dr Kemen Austin

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This paper examines the “climate finance” required to deliver the forest-based mitigation contained in Nationally Determined Contributions (NDC’s) to the Paris Agreement. The focus is on NDC’s in 2030, and the analysis utilizes the GTM, a model that simulates global timber markets thereby accounting for anticipated behavior by landowners. A reference case shows where NDC’s will and will not be met based on projected market activity. Subsequent scenarios estimate the amount of climate finance required to meet NDC’s in each region, highlighting regional differences in implied carbon prices, and examines a globally coordinated approach that highlights the advantage of a market-driven approach to providing forest-based mitigation.

The paper makes a useful contribution to discussions regarding the potential for forest-based climate mitigation, especially by highlighting cost-differentials across regions and the potential returns to global coordination through global trade in forest carbon benefits. Most importantly, it evaluates nation level NDCs in the context of global markets for timber and land use, a critical step toward evaluating their practicality.

I find the analysis competently executed using sound methodology and the paper well-written. To improve the paper, the authors could:

1) Define “climate finance” as they apply it at first mention in the paper.

2) Provide insights into the role of timber demand growth in their projections with some sensitivity analysis—e.g., consider the results under no demand growth.

3) Given the evidently low cost of forest-based mitigation in some countries, discuss further opportunities for expanding the forest contributions and improving the cost efficiency of NDCs for some countries.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
This study provides a valuable complement to study [1], offering novel and significant insights. While study [1] takes a longer-term perspective, demonstrating that global mitigation of 0.6–6.0 GtCO2/year can be achieved under carbon price scenarios of $5–$100/tCO2 at costs ranging from $2 billion to $393 billion per year, the current study estimates that, with global coordination, 4.6 GtCO2/year can be mitigated at an annual cost of $76 billion. This suggests that optimizing climate finance allocation could yield mitigation benefits approximately three times greater than in a baseline scenario with regional or local NDC targets. While these findings are important, they are not entirely unexpected. The authors acknowledge that the practical applicability of their results is constrained by the high transaction costs associated with establishing a globally coordinated climate finance system for forests.

I believe the results are sound and provide novel findings that have not been previously reported in the literature. However, the presentation of the model and scenarios needs refinement. The description of the methodology and the underlying GTM

is better explained in study [1], and as a result, the reader may feel that not all assumptions and underlying parameters are sufficiently detailed in the current paper.

Additionally, the code required to reproduce the results has not been uploaded and is not accessible through the provided link. The scenarios and their assumptions need to be explained in much greater detail to ensure full transparency and reproducibility. Even developing further scenarios should be considered.

Furthermore, the analysis appears to be static, focusing solely on the outcome for the year 2030 without addressing the dynamic evolution beyond this point. How are rotation periods and rising timber prices accounted for? The carbon payments are assumed to be increasing at 3%, how plausible is this assumption? The authors should provide a deeper discussion regarding the sensitivity of results with respect to the assumption of parameters. Additionally, it is puzzling that carbon fertilization forest management is considered an option in the GTM but seemingly not included as a cost to forest owners. More broadly, the concept of opportunity costs associated with alternative forest management practices is neither discussed nor integrated into the analysis.

Incorporating a dynamic perspective on the climate forest market would significantly strengthen the study's contribution.

Ultimately, the paper and its assumed scenarios should be fully self-contained and comprehensible without requiring reference to previous studies.

Reference

[1] K. G. Austin, J. S. Baker, B. L. Sohngen, C. M. Wade, A. Daigneault, S. B. Ohrel, S. Ragnauth,
425 A. Bean, The economic costs of planting, preserving, and managing the world's forests to
426 mitigate climate change. Nat Commun 11, 5946 (2020).

(Remarks on code availability)
I could not find the codes under the link provided. I could also find codes related to previous papers.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have thoroughly addressed the concerns I raised in my original review. I recommend accepting the manuscript in its current form.

(Remarks on code availability)
While I have not attempted to reproduce the results, it appears that the authors have provided the code and datasets necessary for reproducing findings.

Reviewer #2

(Remarks to the Author)
The manuscript is improved after revision and my concerns have been properly addressed.

(Remarks on code availability)
A Readme.doc with instructions how to run the code for replication purpose is not available and should be added.

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Targeting climate finance for global forests - Response to editor and reviewers' comments

Reviewer #1 (Remarks to the Author):

This paper examines the "climate finance" required to deliver the forest-based mitigation contained in Nationally Determined Contributions (NDC's) to the Paris Agreement. The focus is on NDC's in 2030, and the analysis utilizes the GTM, a model that simulates global timber markets thereby accounting for anticipated behavior by landowners. A reference case shows where NDC's will and will not be met based on projected market activity. Subsequent scenarios estimate the amount of climate finance required to meet NDC's in each region, highlighting regional differences in implied carbon prices, and examines a globally coordinated approach that highlights the advantage of a market-driven approach to providing forest-based mitigation.

The paper makes a useful contribution to discussions regarding the potential for forest-based climate mitigation, especially by highlighting cost-differentials across regions and the potential returns to global coordination through global trade in forest carbon benefits. Most importantly, it evaluates nation level NDCs in the context of global markets for timber and land use, a critical step toward evaluating their practicality.

We thank the reviewer for highlighting the broader significance of this analysis, and the positive feedback.

I find the analysis competently executed using sound methodology and the paper well-written. To improve the paper, the authors could:

1) Define "climate finance" as they apply it at first mention in the paper.

We agree that this is valuable addition, and addressed this by adding a specific definition of climate finance to the introduction section of the paper (L55 - 60):

The current scale of climate finance for forests, estimated at $2.3 billion annually (9), is at least an order of magnitude smaller than what is needed to achieve mitigation compatible with a 1.5°C pathway (10, 11). Climate finance broadly refers to funding from public, private, or other sources, used to support climate change mitigation or adaptation actions (12, 13). It can take forms ranging from alignment of public expenditures with climate targets, such as subsidy reform, increases in public funding, such as domestic budget allocations and bilateral or multilateral finance, and mobilization of private finance via mechanisms such as carbon pricing or climate risk disclosure (14).

12. UNFCCC. “Introduction to Climate Finance.” United Nations Framework Convention on Climate Change, Bonn, Germany. https://unfccc.int/topics/introduction-to-climate-finance.

13. K. Zhou, D. Midkiff, R. Yin, H. Zhang, Carbon finance and funding for forest sector climate solutions: a review and synthesis of the principles, policies, and practices. *Front. Environ. Sci.* **12** (2024).

14. M. Caldwell, N. Alayza, G. Larsen, “Paying for the Paris Agreement” (Report, World Resources Institute, Washington, DC, 2022).

2) Provide insights into the role of timber demand growth in their projections with some sensitivity analysis—e.g., consider the results under no demand growth.

Based on this comment and a similar request from Review #2, we ran model sensitivity tests examining how changes in timber demand growth, carbon price increases, and CO2 fertilization would impact our resulting projections. Our approach and results are presented in the main text (L182 - 201) and as a supplementary figure S1:

To assess the robustness of our findings, we examined the impacts of three alternative assumptions: slower forest growth due to absence of CO_2 fertilization, lower future increases in timber demand (10% lower relative to the main findings over 2025 - 2055), and slower increases in carbon prices (1% as opposed to 3% growth). Overall, the results presented in the paper are relatively insensitive to these alternative assumptions (Figure S1). This is due in large part to the time horizon of this study, as the anticipated impacts of these changes in carbon price growth, timber demand, and CO_2 fertilization are projected to manifest over many decades and have larger implications in mid-century and beyond.

We estimate that mitigation outcomes per dollar invested are most sensitive to assumptions about CO_2 fertilization. Under a globally coordinated $70 billion investment scenario, excluding assumptions of CO_2 fertilization leads to a 10% lower projection of global mitigation in 2030 (Figure S1, grey line). This result is expected, as investments that enhance forest productivity are aided by anticipated increases in tree growth due to CO_2 fertilization. In contrast, lowering global timber demand results in a 5% increase in projected mitigation per dollar invested under the $70 billion investment case in the short-term (Figure S1, yellow line). Finally, changes in the growth in carbon prices have a negligible impact in the short term; under the same $70 billion scenario, results remain

effectively unchanged (Figure S1, orange line). Price dynamics may play a more significant role in medium- to long-term projections by influencing forward-looking investment decisions.

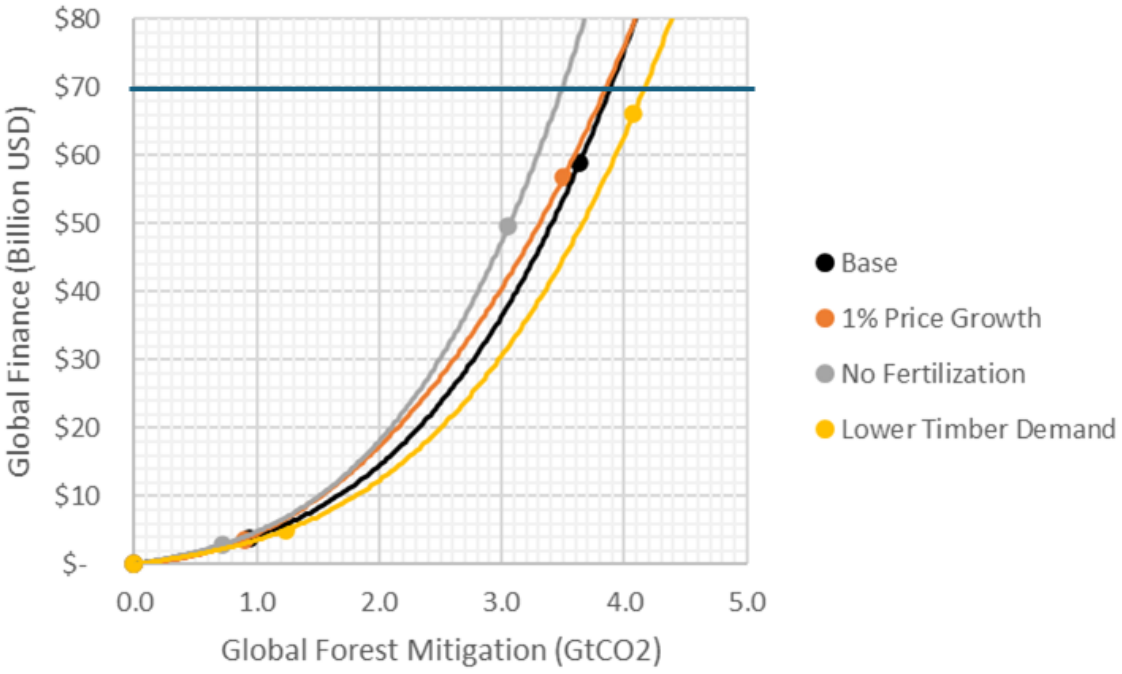


Figure S1. Quantity of mitigation from forest-based mitigation actions with different levels of aggregate global climate finance, projected by the Global Timber Model. The "Base" case is the scenario tested and reported in this study, which assumes that CO2 fertilization occurs consistent with an RCP 4.5 scenario, as well as a 3% growth rate in carbon prices. The assumptions in each of the other three scenarios are held the same, except for a single modification. The "1% Price Growth" scenario assumes that carbon prices will increase at a slower rate of 1%. The "No fertilization" scenario assumes no impacts of CO2 fertilization on tree growth. Finally, the "Lower Timber Demand" scenario assumes 10% lower timber demand over 2025-2055.

3) Given the evidently low cost of forest-based mitigation in some countries, discuss further opportunities for expanding the forest contributions and improving the cost efficiency of NDCs for some countries.

This is important, and we added additional clarity and emphasis on opportunities for increasing ambition in forest-based NDCs, to the Results section (L152-166):

These cost efficiencies are the product of a geographic shift in mitigation under the Global Coordination scenario, with large increases in low-cost mitigation in some regions, and decreases in very high-cost mitigation in other regions, relative to regional NDC targets (Figure 2). Under the Global Coordination scenario, we project that Canada, Russia, China, Central America, and Oceania begin to take advantage of low-cost abatement opportunities that are not reflected in the in the first generation of their NDCs. Together, these five regions are projected to reach 0.3 Gt CO_2 of mitigation in 2030, requiring finance (either domestic, international or both) of $5 billion annually. We also project substantially more mitigation in Sub Saharan Africa under the Global Coordination scenario, reaching net mitigation of 1.2 Gt CO_2 with climate finance of $21 Billion annually in 2030, Brazil, reaching 0.9 Gt CO_2 with climate finance of $16 Billion annually, and the rest of South America, reaching 0.5 Gt CO_2 with climate finance of $9 Billion annually. On the other hand, we project somewhat less mitigation in SE Asia, Japan, East Asia, and Northern Africa and the Middle East, relative to these regions' aggregate NDC commitments, due to the comparatively higher cost of mitigation in these locations.

Reviewer #2 (Remarks to the Author):

This study provides a valuable complement to study [1], offering novel and significant insights. While study [1] takes a longer-term perspective, demonstrating that global mitigation of 0.6–6.0 GtCO2/year can be achieved under carbon price scenarios of $5–$100/tCO2 at costs ranging from $2 billion to $393 billion per year, the current study estimates that, with global coordination, 4.6 GtCO2/year can be mitigated at an annual cost of $76 billion. This suggests that optimizing climate finance allocation could yield mitigation benefits approximately three times greater than in a baseline scenario with regional or local NDC targets. While these findings are important, they are not entirely unexpected. The authors acknowledge that the practical applicability of their results is constrained by the high transaction costs associated with establishing a globally coordinated climate finance system for forests.

We thank the reviewer for the thoughtful feedback and consideration of the value of the study.

I believe the results are sound and provide novel findings that have not been previously reported in the literature. However, the presentation of the model and scenarios needs

refinement. The description of the methodology and the underlying GTM is better explained in study [1], and as a result, the reader may feel that not all assumptions and underlying parameters are sufficiently detailed in the current paper.

Thank you, we agree and added substantial additional detail regarding the model design and calibration, forest type representation, optimization approach, timber demand representation, land supply functions, and carbon stock quantification. These details are provided in the Methods section (L365 - 485) and are sufficiently extensive that we do not copy them here.

Additionally, the code required to reproduce the results has not been uploaded and is not accessible through the provided link. The scenarios and their assumptions need to be explained in much greater detail to ensure full transparency and reproducibility. Even developing further scenarios should be considered.

We apologize for this oversight. The link to the code has been corrected and is now accessible for review at: https://www.dropbox.com/scl/fo/igypjibs4vmkrazzc9yqx/AP6IWx9c53uJIVl2NvQS6oQ?rlkey=b7yk95q53p76yjmxu08tyhdxk&st=2gmo1a5u&dl=0

This link is also available via the Global Timber Model Code Repository hosted by the Ohio State University at: https://u.osu.edu/forest/code-repository/

We also added new detail on the model assumptions, in particular on CO2 fertilization (L434-448) and the development of the Scenarios (L482 - 509).

Finally, we added several new sensitivity test, described in response to the next comment.

Furthermore, the analysis appears to be static, focusing solely on the outcome for the year 2030 without addressing the dynamic evolution beyond this point. How are rotation periods and rising timber prices accounted for? The carbon payments are assumed to be increasing at 3%, how plausible is this assumption? The authors should provide a deeper discussion regarding the sensitivity of results with respect to the assumption of parameters. Additionally, it is puzzling that carbon fertilization forest management is considered an option in the GTM but seemingly not included as a cost to forest owners. More broadly, the concept of opportunity costs associated with alternative forest management practices is neither discussed nor integrated into the analysis.

We focus on the near term 2030 targets as a specific application and knowledge gap to be filled, to inform NDC refinement/ updating and climate finance roadmap development – steps that are currently underway. However, we agree that there is also value in examining dynamics mid-

century and beyond, especially given the ongoing development and refinement of mid-century strategies aligned with Paris Agreement targets. We add a note on this to the Discussion Section (L269-276):

Further, the results presented here focus on short-term mitigation opportunities aligned with the 2030 NDC targets. A similar analysis could be expanded to include medium- (2050) and long-term (2100) perspectives, which would allow for evaluation of the cost efficiency of a broader set of national targets and could inform their design and refinement. Indeed, near-term price signals associated with climate mitigation finance may drive further action, leading to changes in forest management and afforestation/reforestation and greater sequestration after 2030. This is especially important for forests, where current investments take time to yield results due to the growth rate of forests.

We also very much agree with the reviewer's points that many of the parameters and assumptions in the model have some level of uncertainty. We clarify how we represent these parameters – timber demand and resulting prices, carbon price growth, CO2 fertilization – in the methods section. We also add new sensitivity tests, something that Review #1 also requested, examining how changes in timber demand growth, carbon price increases, and CO2 fertilization would impact our resulting projections. Our approach and results are presented in the main text (L182 - 201) and as a supplementary figure S1:

To assess the robustness of our findings, we examined the impacts of three alternative assumptions: slower forest growth due to absence of CO_2 fertilization, lower future increases in timber demand (10% lower relative to the main findings over 2025 - 2055), and slower increases in carbon prices (1% as opposed to 3% growth). Overall, the results presented in the paper are relatively insensitive to these alternative assumptions (Figure S1). This is due in large part to the time horizon of this study, as the anticipated impacts of these changes in carbon price growth, timber demand, and CO_2 fertilization are projected to manifest over many decades and have larger implications in mid-century and beyond.

We estimate that mitigation outcomes per dollar invested are most sensitive to assumptions about CO_2 fertilization. Under a globally coordinated $70 billion investment scenario, excluding assumptions of CO_2 fertilization leads to a 10% lower projection of global mitigation in 2030 (Figure S1, grey line). This result is expected, as investments that enhance forest productivity are aided by anticipated increases in tree growth due to CO_2 fertilization. In contrast, lowering global timber demand results in a 5% increase in projected mitigation per dollar

invested under the $70 billion investment case in the short-term (Figure S1, yellow line). Finally, changes in the growth in carbon prices have a negligible impact in the short term; under the same $70 billion scenario, results remain effectively unchanged (Figure S1, orange line). Price dynamics may play a more significant role in medium- to long-term projections by influencing forward-looking investment decisions.

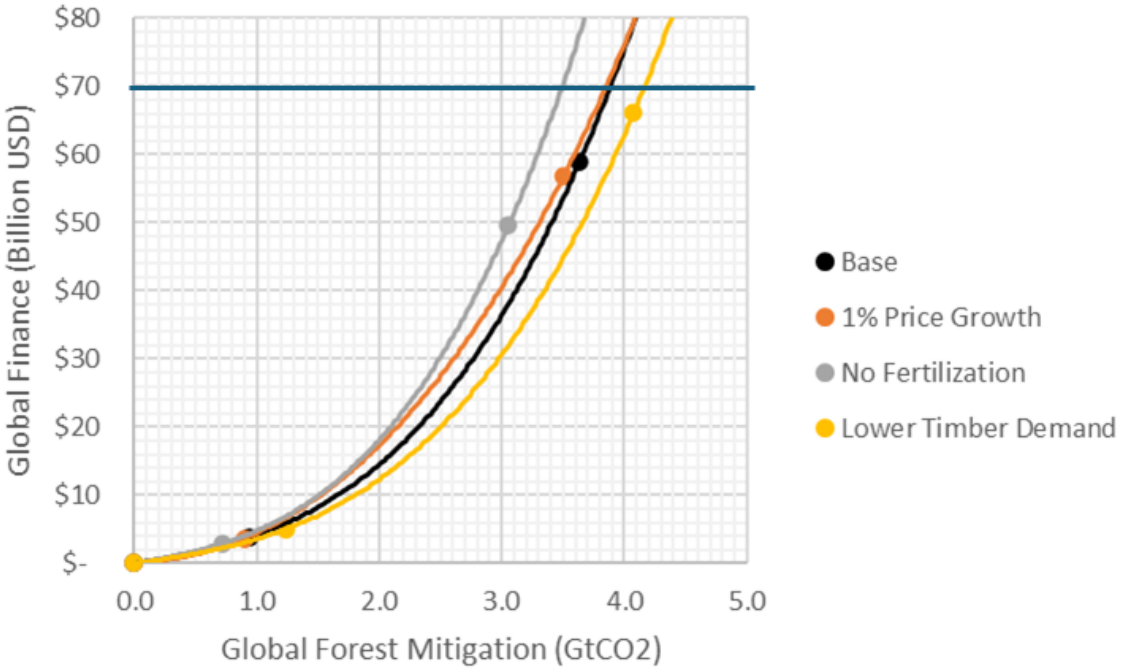


Figure S1. Quantity of mitigation from forest-based mitigation actions with different levels of aggregate global climate finance, projected by the Global Timber Model. The "Base" case is the scenario tested and reported in this study, which assumes that CO2 fertilization occurs consistent with an RCP 4.5 scenario, as well as a 3% growth rate in carbon prices. The assumptions in each of the other three scenarios are held the same, except for a single modification. The "1% Price Growth" scenario assumes that carbon prices will increase at a slower rate of 1%. The "No fertilization" scenario assumes no impacts of CO2 fertilization on tree growth. Finally, the "Lower Timber Demand" scenario assumes 10% lower timber demand over 2025-2055.

Since our main focus is on assessing the effects of carbon payments relative to a base case without finance, we examined these incentives in the low-demand scenario. In the short term,

lower timber demand leads to a slight increase in mitigation per dollar invested (i.e., lower timber demand reduces the costs of forest-based carbon sequestration).

Finally, we clarify that our modelling approach does include consideration of opportunity costs of land. In GTM, countries can respond to carbon payment incentives for forest mitigation through multiple pathways: by altering forest management practices (e.g., adjusting harvest volumes or rotation lengths), by afforesting currently non-forested areas, or by converting unmanaged forests into managed forests. Each country selects its optimal mix of these strategies based on current forest conditions, projected market demand for timber products, expected revenues from carbon payments, and associated production and land conversion costs.

Because countries are linked through international timber markets, their decisions are interdependent—actions taken by one country influence global market prices and, in turn, the decisions of others. The model captures these dynamics by maximizing the overall welfare of the global timber sector, including both producers and consumers. Within this framework, countries also account for the opportunity cost of land use. For example, when considering whether to convert non-forested land to forest, they weigh the potential carbon and timber revenues against the forgone income from the land's existing use. The substantially expanded Methods Section clarifies this.

Incorporating a dynamic perspective on the climate forest market would significantly strengthen the study's contribution.

We apologize for lack of clarity regarding the model representation of dynamic factors, as well as opportunity costs. The analysis is not static, which is one of the key advantages of using a dynamic optimization model like GTM. This model provides important insights into forest mitigation potential by integrating both market and biophysical dynamics. Rotations and management strategies are endogenous within the model and respond to carbon payments tied to carbon stored in forests and long-term timber products. These changes can increase forest carbon sequestration per hectare, but they are only implemented when the revenue lost from reducing harvests is equal to or less than the revenue gained from the additional carbon storage. Additionally, an expected increase in natural fertilization could lead to higher forest productivity, which may drive more investment. In such cases, more carbon could be sequestered per dollar invested compared to a scenario without fertilization. This dynamic highlights how both biophysical processes and economic incentives interact within the model to influence forest management decisions and mitigation outcomes. The substantially expanded Methods Section clarifies this.

Ultimately, the paper and its assumed scenarios should be fully self-contained and comprehensible without requiring reference to previous studies.

Reference

[1] K. G. Austin, J. S. Baker, B. L. Sohngen, C. M. Wade, A. Daigneault, S. B. Ohrel, S. Ragnauth,
425 A. Bean, The economic costs of planting, preserving, and managing the world's forests to
426 mitigate climate change. Nat Commun 11, 5946 (2020).

Reviewer #2 (Remarks on code availability):

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